

Mounika Imadabathuni

Portfolio

Email : imadabathunim@gmail.com

Mobile : +91-9392648082

EDUCATION

BVRIT Hyderabad College of Engineering for Women <i>Bachelor of Technology – Electronics and Communication Engineering; CGPA: 8.9/10 (Till Date)</i>	Hyderabad, India 2023 – 2027
Sri Chaitanya Junior College <i>Intermediate (Class XII), MPC; Percentage: 95.40%</i>	Hyderabad, India 2021 – 2023
Sri Vijnana Vihara EM School <i>Secondary School (Class X); Percentage: 97.80%</i>	Vijayawada, India 2020 – 2021

EXPERIENCE

Research Intern, AESICD Lab <i>Indian Institute of Technology Hyderabad (IITH)</i>	Hyderabad, India Dec 2025 – Jun 2026
- Working on AI-enabled, energy-efficient, lightweight BMS-on-chip architectures for advanced energy storage systems. - Implemented RTL to GDS flow of a QRD block using Cadence and Synopsys tools on the TSMC 28nm PDK for integration into an AI-enabled Battery Management SoC.	

PROJECTS

Battery Management SoC (BMS) – QRD Physical Design - Implemented the complete RTL-to-GDSII physical implementation flow of the QRD block using TSMC 28nm HPC technology. Tech Stack: Verilog, SpyGlass, Xcelium, Cadence Genus, Innovus, Tempus, Voltus, Calibre, TSMC 28nm HPC. Contribution: Performed RTL linting, synthesis, floorplanning, placement, CTS, routing, timing closure, DRC, and post-layout verification.
Autonomous Low Power Sensor Hub (ALSH) Lead Synthesis - Developed an energy-efficient Always-On-Network (AON) hardware gatekeeper to filter sensor noise using a Moving Average Algorithm. Tech Stack: TSMC 28nm, Cadence Genus Contribution: Led the synthesis flow and optimized the design for low-power operation.

SKILLS SUMMARY

Languages & Tools: Verilog, SystemVerilog, C++, Cadence Genus, Innovus, Xcelium, PrimeTime, VCS, Calibre, Git
Platforms & Environments: Linux
Domain Knowledge: RTL Design, ASIC Design Flow, Physical Design, Static Timing Analysis (STA)
Soft Skills: Team Work, Communication, Leadership

CERTIFICATIONS

VLSI Foundations <i>HCLTech</i>	View
Semiconductor Fabrication <i>Purdue University</i>	View
Research Internship <i>Indian Institute of Technology Hyderabad (AESICD Lab)</i>	View

ACHIEVEMENTS

NASA Space Apps Challenge 2025: Secured National Top 5.
Smart India Hackathon (SIH) 2025 (Hardware): Winner.

VOLUNTEER EXPERIENCE

President – Rotaract Club of BVRIT Hyderabad: Leading a team of 108 students to organize social and technical initiatives, improving outreach by 25%.
--